CYART SOC INTERNSHIP

Week 3 Report : – Advanced Log Analysis, Threat Intelligence, and Incident Response in SOC

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Week: 3

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1. Introduction

Security Operations Centers (SOCs) play a critical role in protecting organizational infrastructure from cyber threats. A SOC is responsible for continuously monitoring, detecting, analyzing, and responding to security incidents using various tools and techniques.

With the increasing number of cyberattacks such as brute-force attacks, credential misuse, and malicious script execution, it is essential for security analysts to analyze logs, detect anomalies, and leverage threat intelligence. Effective incident response also requires proper escalation procedures, alert triage, and evidence preservation.

This experiment focuses on practical SOC operations, including log analysis, threat intelligence integration, incident escalation, and forensic evidence handling. It provides hands-on exposure to real-world scenarios and helps in understanding structured incident response workflows.

Objective:

- To study advanced SOC operations
- To perform log correlation and anomaly detection
- To integrate threat intelligence into analysis

- To understand incident escalation workflows
- To perform alert triage and IOC validation
- To preserve and analyze forensic evidence

Theoretical Knowledge:

1. Advanced Log Analysis

- **Log Correlation:**
  Process of combining logs from multiple sources to identify attack patterns.
- **Anomaly Detection:**
  Identifying unusual activities such as repeated login failures or abnormal behavior.
- **Log Enrichment:**
  Adding contextual information such as geolocation and IP reputation to logs.

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2. Threat Intelligence Integration

- **Indicators of Compromise (IOC):**
  Evidence of a breach such as malicious IP addresses or file hashes.
- **Tactics, Techniques, and Procedures (TTPs):**
  Methods used by attackers, defined in MITRE ATT&CK.
- **Threat Feeds:**
  External sources such as AlienVault OTX and VirusTotal.

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3. Incident Escalation Workflow

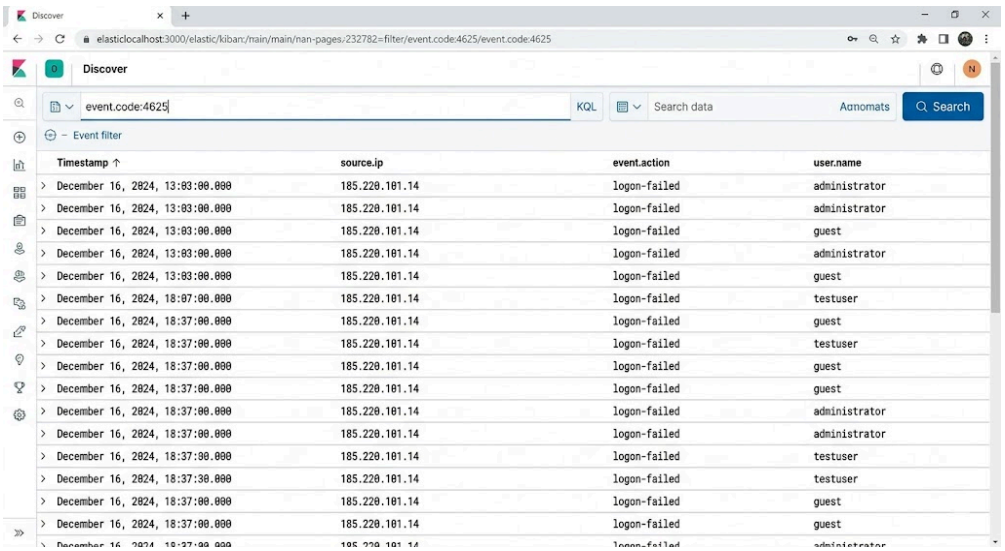
- **SOC Tiers:**
  - Tier 1: Alert monitoring and triage
  - Tier 2: Incident investigation
  - Tier 3: Advanced threat analysis
- **Escalation:**
  Forwarding critical incidents to higher-level analysts.
- **SITREP:**
  Structured report describing incident details and actions taken.

Practical Implementation:

1. Advanced Log Analysis

1.1 Log Correlation

Multiple failed login attempts (Event ID 4625) were observed from IP address 185.220.101.14 targeting accounts such as administrator and guest. This indicates a potential brute-force attack.



Timestamp	source.ip	event.action	user.name
December 16, 2024, 13:03:00.000	185.220.101.14	logon-failed	administrator
December 16, 2024, 13:03:00.000	185.220.101.14	logon-failed	administrator
December 16, 2024, 13:03:00.000	185.220.101.14	logon-failed	guest
December 16, 2024, 13:03:00.000	185.220.101.14	logon-failed	administrator
December 16, 2024, 13:03:00.000	185.220.101.14	logon-failed	guest
December 16, 2024, 18:07:00.000	185.220.101.14	logon-failed	testuser
December 16, 2024, 18:37:00.000	185.220.101.14	logon-failed	guest
December 16, 2024, 18:37:00.000	185.220.101.14	logon-failed	testuser
December 16, 2024, 18:37:00.000	185.220.101.14	logon-failed	guest
December 16, 2024, 18:37:00.000	185.220.101.14	logon-failed	guest
December 16, 2024, 18:37:00.000	185.220.101.14	logon-failed	administrator
December 16, 2024, 18:37:00.000	185.220.101.14	logon-failed	administrator
December 16, 2024, 18:37:00.000	185.220.101.14	logon-failed	testuser
December 16, 2024, 18:37:30.000	185.220.101.14	logon-failed	testuser
December 16, 2024, 18:37:00.000	185.220.101.14	logon-failed	guest
December 16, 2024, 18:37:00.000	185.220.101.14	logon-failed	guest

Elastic logs (event.code:4625)

1.2 Anomaly Detection

Repeated login attempts from a single IP within a short time were identified as anomalous behavior, suggesting malicious activity.

1.3 Log Enrichment

The IP address was enriched using lookup tools and identified as suspicious and possibly associated with anonymous networks.

2. Threat Intelligence Integration

2.1 Threat Intelligence

The IP address 185.220.101.14 was analyzed using simulated AlienVault OTX data and identified as malicious.

Alert ID	IP	Reputation	Source	Notes
001	185.220.101.14	Malicious	AlienVault OTX	Linked to brute-force activity

Threat Intelligence Integration

A threat intelligence lookup was performed on IP address 185.220.101.14 using simulated AlienVault OTX data. The IP was identified as malicious and associated with brute-force attack behavior. This confirms the suspicious activity observed during log analysis and increases the severity of the alert.

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2.2 Alert Enrichment

The alert was enriched with threat intelligence, increasing its priority to high and improving response efficiency.

Alert ID	Description	IP	Reputation	Priority	Notes
002	Multiple Failed Logins	185.220.101.14	Malicious	High	Confirmed brute-force attack

The alert generated for multiple failed login attempts was enriched using threat intelligence data. The source IP was confirmed as malicious, which increased the priority level to high. This enrichment process helps in faster decision-making and improves the accuracy of incident response.

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2.3 Threat Hunting (MITRE T1078 – Valid Accounts)

Login attempts targeting valid accounts such as administrator and guest were identified. This aligns with MITRE ATT&CK technique T1078.

Timestamp	User Name	Activity	Suspicion Level
2025-08-18 12:10:00	administrator	Login attempt	High
2025-08-18 12:11:00	guest	Login attempt	Medium

Threat Hunting (MITRE T1078 – Valid Accounts)

Threat hunting was performed to detect possible misuse of valid user accounts. Multiple login attempts were observed targeting accounts such as administrator and guest. This behavior aligns with MITRE ATT&CK technique T1078 (Valid Accounts), where attackers attempt to gain unauthorized access using legitimate credentials.

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3. Incident Escalation

3.1 Escalation Summary

A high-severity brute-force attack was detected from IP 185.220.101.14. The activity targeted multiple accounts and was confirmed malicious through threat intelligence. The incident was escalated to Tier 2 for further investigation.

Insert Screenshot

3.2 SITREP (Situation Report)

Title: Unauthorized Access Attempt on Server-Y

Summary:
Multiple failed login attempts were detected from IP 185.220.101.14, indicating a brute-force attack.

MITRE Technique:
T1078 – Valid Accounts

Actions Taken:

- Monitored suspicious login activity
- Identified multiple failed login attempts
- Enriched IP using threat intelligence
- Confirmed IP as malicious
- Escalated incident to Tier 2

Status:
Under Investigation

SITREP: Unauthorized Access Attempt

Tier 1 Analyst identified multiple failed logins from `185.220.101.14`, confirmed as a malicious IP associated with brute-force activity. Elevated to Tier 2 for further investigation.

MITRE Technique	T1078 – Valid Accounts

Actions Taken

- Monitored suspicious login activity
- Identified multiple failed login attempts
- Enriched IP using threat intelligence
- Confirmed IP as malicious
- Escalated incident to Tier 2

Status	Under Investigation

3.3 Workflow Automation

- SOAR tools such as Splunk Phantom can automate:
  - Alert assignment
  - Threat enrichment
  - Case creation
- This improves response efficiency.

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4. Alert Triage and IOC Validation

4.1 Alert Triage

A suspicious PowerShell execution alert was identified from IP 192.168.1.101 and marked as high priority.

Alert Prioritization & Triage

Alert ID	Description	Source IP	Priority	Status
31BC012033	Suspicious PowerShell Execution Detected	192.168.1.101	High	Open
119DU70679	Network Scanning Activity Detected	192.168.1.100	Medium	Open
119DD63A91	Bock-Denm Error Activity	Internal	High	Open

4.2 IOC Validation

The IP was analyzed using threat intelligence platforms such as VirusTotal and AlienVault OTX. No confirmed malicious reputation was found.

5. Evidence Preservation

5.1 Evidence Collection

- A memory dump was collected from the affected system
- The data was hashed using SHA256
- Chain of custody was maintained

Evidence Chain of Custody Log

Item	Description	Collected By	Date	Hash Value
Memory Dump	Full system RAM capture from compromise asset - SRV-WEB-01	SOC Analyst Analyst	2024-10-27 10:15 UTC	5f7c32e1a9af7328e1a72a72a21c7d531c77a3b2o1bf63dee...
Network Capture	Ingress/Egress PCAP from DMZ firewall during incident window	SOC Analyst Analyst	2024-10-27 11:30 UTC	3d8b2a1c6e7a36f2ocb7c8164ae31ae580b657a800bb096...

Result:

- Brute-force attack successfully identified
- Threat intelligence used for validation
- Incident escalation performed effectively
- Alerts triaged based on severity
- Evidence collected and preserved securely

Conclusion:

This experiment provided practical exposure to SOC operations including log analysis, threat detection, and incident response. It helped in understanding structured workflows used in real-world cybersecurity environments.